Response to the reviewers — manuscript OL-03-2026-11

Manuscript OL-03-2026-11, now titled *Uncoded basic vocabulary in six Sulawesi word lists of the Austronesian Basic Vocabulary Database*.

We are grateful to both reviewers. The first reviewer's comments corrected our classification and terminology and sent us to the literature on Makasar and Tolaki; the second reviewer's highlights told us, point by point, where a reader of the journal would be lost in Sections 2 and 3. We have tried to answer every one of them.

How the revised version relates to the submitted one

At the editor's request, this note says in general terms how the revised version relates to the submitted one; Parts A to C give the detail.

- **What is the same.** The six Sulawesi word lists of the ABVD, with the same 1,357 forms; a classifier that is assessed on forms it has not seen, by cross-validation and with one list held out; and tests of how far it depends on spelling. The limitations, the declaration on the use of artificial intelligence and the open code and data, which the second reviewer singled out, are kept and rewritten.
- **What the paper now claims.** Less than before. The classifier scores each form from its written form and its meaning only: the two inputs at the level of the list (the identity of the source list and a coverage variable) are removed, so its AUC is 0.727 rather than the submitted 0.763 (0.672 from the written form alone), and it is presented as a ranking aid rather than a detector of substrate vocabulary. The profile is restated without the claim of fewer canonical prefixes, and the result is found to depend partly on how the glottal stop is written; the submitted statement that it is robust rather than an orthographic artifact is withdrawn. The negative result about a shared layer is replaced by a statement of what a permutation test measures, and the paper makes no claim about a shared layer either way; the application to sixteen further lists is reduced to one paragraph without a geographic claim; and two new subsections on Makasar and Tolaki answer the first reviewer's questions with counts from the data. The title and the abstract have changed accordingly. Many of these changes correct errors of our own, which we found when we re-derived every number of the submitted text from the raw ABVD files and of which we told the editors before revising; the editor has agreed that we correct them in this revision. Part C lists them.
- **Terms.** One label with two values, *coded* and *uncoded*, replaces the two label sets of the submitted version and the "residuals" built on them, so the Tolaki list released for inspection has 134 items rather than the 114 of the submitted Table 1; "phonological fingerprint" becomes *profile*; "false positive" is no longer used, and the internal experiment codes are gone from the text.
- **Sections.**
    - Section 1 is rewritten. It now gives the literature on the subgroups of Sulawesi and on Makasar that the first reviewer named, and defines *coded*, *uncoded* and *profile*.
    - In Section 2, the rule-based baseline of submitted Section 2.2 has become the label of Section 2.2, with one definition and without the Proto-Austronesian cross-check and the loanword filter; the classification of submitted Section 2.3 is explained in words in Sections 2.3 to 2.5 (inputs, classifier, assessment), without the two-model design; the consensus of submitted Section 2.4 and the clustering of submitted Section 2.5 are removed; Section 2.6 is new and describes the three checks of what "uncoded" means (Makasar against Proto-Malayo-Polynesian, Tolaki against its subgroup, similarity across

lists) and the spelling tests. The list of eight meanings said to be residual in five or six lists (given in submitted Section 2.2 and again in submitted Section 3.1) is not kept: it counted forms rather than lists (Part C).
    - In Section 3, Section 3.1 (Table 1) corresponds to submitted Section 3.1 under the one label; Sections 3.2 (Makasar) and 3.3 (Tolaki) are new; Section 3.4 corresponds to submitted Section 3.2 (cross-validation, one list held out); Section 3.5 takes the place of submitted Section 3.5 (SHAP), of submitted Section 3.6 (consensus) and of the remarks on the numerals 'Fifty' and 'Twenty' in submitted Section 3.7; Section 3.6 takes up the spelling test of submitted Section 3.4 (digraphs, word length) and adds the tests on the glottal stop; Section 3.7 replaces the clustering of submitted Section 3.7 with the permutation test; Section 3.8 reduces submitted Section 3.8 to one paragraph. The feature ablation of submitted Section 3.3 is not kept as a section: the model of the revision is the variant without both list-level inputs, the last row of submitted Table 4.
    - Section 4 is shorter: six paragraphs, each headed by its topic (what the profile is, what the classifier is for, Makasar, Tolaki, similarity across lists, limitations). The Makasar paragraph is new; the subsection of the submitted Discussion on Javanese script is withdrawn; its subsection on parallel innovation gives way to the paragraph on similarity across lists; and its comparison with traditional methods, which in part rested on the consensus figures, is withdrawn, the role of the classifier being stated in the paragraph on what the classifier is for. Section 5 is rewritten: four statements, then directions for further work; the submitted paragraphs on Javanese script and on the two-model design as a template are dropped.
- **Figures and tables.** Figure 1 replaces the submitted Figure 1: a bar chart of the inputs of the revised model in place of the plot of the submitted 27-input model. Submitted Figures 2 to 4 are removed; the comparison of label and model in submitted Figure 2 is given, out of sample, as Table 6. Figures 2 to 5 are new (the workflow, the shares of Table 2, the AUC columns of Table 4, the written glottal mark by list). Table 1 corresponds to submitted Table 1 under the one label, and Tables 3 and 4 to submitted Tables 2 and 3, rebuilt for the model of the revision; submitted Tables 4 and 5 are removed; Tables 2, 5, 7 and 8 are new.
- **Tracked changes.** A copy of the article with tracked changes against the submitted text has been supplied to the editors. Because Sections 2 and 3 were largely rewritten, much of its markup consists of replaced blocks, and this note is the better guide.

The answers are given as bullets, as the Managing Editor advised: one for every point of the two reports (Parts A and B, in the order of the reports; the labels R1-1 to R1-13 and R2-1 to R2-24 are ours), with more detail where a point is substantive; Part C lists the corrections by the authors. Section, table and figure numbers are those of the revised manuscript unless "submitted" is said; the bold headings give the reviewer's point and so refer to the submitted version. This document carries no author names. The DOI of the data release and the address of the code repository name the authors; they are given to the editors in the cover letter and are withheld here, as they are in the article file.

Part A. Reviewer 1

- **Opening remark — the computational terminology is hard to follow; the caption "SHAP beeswarm plot for Model B" is the example given.** We agree. The second reviewer's highlights concern the same terms and are answered in Part B; for the figure behind that caption:
    - Figure 1 is now a plain bar chart of the mean absolute contribution of each input to the predictions of the full model, with the inputs named by what the code computes and, for each input, an arrow for its direction or a dot where there is no clear direction, explained

in the legend; the software title, the raw variable names and the caption's sign error are gone.
    - Section 2.4 says what the bars and the arrows measure, and Section 3.5 says what the figure is and is not: a description of the classifier, not of the vocabulary.
- **R1-1 — "resist reconstruction to any proto-form" is too strong.** Accepted.
    - The wording is now "cannot be assigned to a higher-level protoform" (first sentence of Section 1); the other sentences that carried the old wording, in the abstract and in the introduction, are gone with the rewriting.
    - The point reaches further than wording: Section 3.3 reports that 70.9 percent of Tolaki's uncoded forms have a same-meaning look-alike in Proto-Bungku-Tolaki or in one of 42 other Bungku-Tolaki lists (10.5 percent expected by chance; 95.9 percent for the coded Tolaki forms). The text states that this is a mechanical screen across lists that are almost all one compiler's, not a set of etymologies, and that recurrence within a subgroup says nothing about origin — the reviewer's own example of a word borrowed by the speakers of a low-level proto-language is given as the reason.
- **R1-2 — Makasar: how does the published 62 percent "open to investigation" fit our figure, and what does the study add?** The new Section 3.2 and Table 2 answer this.
    - The Introduction now gives the literature on Makasar that the reviewer names: its divergence within South Sulawesi (Mills 1975:491; Sirk 1989:71), the substratum raised and left open (Bulbeck 1992:512–13), and the 62 percent of Bulbeck, Pasqua and Di Lello (2000:103).
    - From ABVD, per meaning (201 meanings with a PMP entry), the Makasar form is in the PMP entry's cognate set for 39.3 percent of meanings (Wilson 95 percent interval 32.8 to 46.2), coded in another set for 25.9 percent, and uncoded for 34.8 percent; Bugis 48.3 / 28.4 / 23.4, Sa'dan Toraja 48.0 / 33.0 / 19.0. We name the first class by what is counted (a form in the PMP entry's set): it is a lower bound on the share that ABVD would count as retained, not retention itself, since an uncoded meaning cannot share a set with the PMP entry whatever its origin.
    - What distinguishes Makasar from its relatives in these data is mainly one fact — more uncoded meanings (exact McNemar $p = 0.006$ against Bugis, 0.0001 against Sa'dan Toraja). Its lower share in the PMP set is that fact counted again: on the meanings coded in both lists the shares are practically equal (70 against 69 of 110; 70 against 71 of 115). Its smaller share of common sets with each relative is that fact in part only: on the meanings coded in both lists the shares are 76.3 percent with Bugis and 70.1 with Sa'dan Toraja, against 81.0 between Bugis and Sa'dan Toraja (no test was made of that difference).
    - A look-alike screen finds a PMP look-alike for 2 of 75 uncoded Makasar forms (about one expected by chance; the screen's sensitivity on forms ABVD itself places in the PMP set is 38 percent, an upper bound for cognates the coders missed) and a Bugis or Sa'dan Toraja look-alike for 10 of 80 (chance 3.8 percent; 90 of 136 among coded forms); the uncoded forms of Bugis and of Sa'dan Toraja resemble their relatives' forms at similar rates (8 of 62 and 8 of 45). The screen therefore does not favour the reading that the uncoded forms are merely uncoded cognates; it leaves open what they are.
    - Section 3.2 traces the published figure from Bulbeck, Pasqua and Di Lello (2000:103) back to Bellwood (1997:115). The 38 percent is Blust's lexicostatistical count against a 200-item PMP list as reported by Bellwood, beside a Western Malayo-Polynesian mean of 41 percent; ABVD's PMP list is Blust's too, so our 39.3 percent is a count of the same kind in the same tradition on another Makasar word list. The manuscript says that their

agreement is not independent confirmation and that in its source the figure is not presented as low. Sirk's Table 1 (45 / 42 / 60 for Makasar–Bugis, Makasar–Sa'dan, Bugis–Sa'dan) is given as a more independent comparison beside ABVD's 41.4 / 39.2 / 53.1.

  - The 62 percent "open to investigation" is addressed in Section 3.2 as the complement of a retention count: it contains forms with cognates outside the PMP set as well as forms with none, and so cannot be set against the uncoded share of Table 1.
  - The manuscript states that the data are compatible with the lexical divergence reported in the literature, that they do not establish it, and that they say nothing about its cause. The 70 uncoded Makasar meanings, with the Bugis and Sa'dan Toraja forms and the PMP entry, are in the released files for anyone able to judge them.
  - Three smaller points. The report gives our earlier figure as 30.1 percent; the submitted Table 1 printed 30.9 percent. Mills's statement of the divergence is on p. 491 of the dissertation, with its chart on p. 492 (the pages cited in the report hold cognate sets), and the text cites p. 491. We cite Bulbeck, Pasqua and Di Lello with the year 2000, the volume year of *Asian Perspectives* 39, and van den Berg's paper on the demise of focus with 1996; the report has 2001 and 1989.

- **R1-3 — the cited source lists eleven subgroups; Sneddon 1993 could be cited.** Accepted. Section 1 now says that Blust (2013:82) lists eleven microgroups and that Sneddon (1993:2) counts nine plus Banggai as a group of one; the word "primary subgroups" is dropped, since neither source uses it, and the edition cited is the one consulted (2013).

- **R1-4 — Celebic is a supergroup.** Accepted. The sentence that named Celebic beside South Sulawesi and Muna-Buton is gone. Section 1 now names the four groups that our six lists represent (South Sulawesi, Muna-Buton, Wotu-Wolio and Bungku-Tolaki) and describes Celebic as a supergroup proposed by van den Berg (1996) and developed by Mead (2003), in whose version South Sulawesi lies outside it and some branches are called tentative. Because three of our four groups fall inside it, Section 3.7 says that the four are not four independent witnesses.

- **R1-5 — "parallel innovation" is a technical term.** Accepted; the term is gone, as is the subsection it headed.

  - We have not adopted "independent innovations" in its place, because the test behind the submitted negative result did not test that hypothesis (it compared a mean over twenty meanings with single draws of mostly coded vocabulary).
  - The permutation test that replaces it (Section 3.7) finds that, among uncoded forms, same-meaning forms are slightly more alike across lists than different-meaning forms (none of 10,000 permutations; four pairs within 0.34 against 0.5 expected; the coded forms, run as a positive control, show 472 against 8.5). The test asks only about the uncoded forms: shared vocabulary that ABVD's editors recognised carries a set number and sits in the coded class, which the test does not examine beyond that control. The paper therefore makes no claim about a shared layer, positive or negative.

- **R1-6 — "three subgroups" but four listed; grouping; spelling; abbreviation.** Accepted on every point. The four subgroups are given as the reviewer gives them: Muna-Buton (Muna), Wotu-Wolio (Wolio), South Sulawesi (Bugis, Makasar, Sa'dan Toraja) and Bungku-Tolaki (Tolaki); Sa'dan Toraja is nowhere Celebic. We use *Makasar* and *Sa'dan Toraja*, and Section 2.1 says which ABVD list the Sa'dan Toraja data come from (list 226, "Tae' (S.Toraja)", from van der Veen 1940). The abbreviation of Bolaang Mongondow disappears with the submitted table and figure of the sixteen further languages (Part C).

- **R1-7 — "under-documentation" is unclear.** The word is withdrawn (Section 3.3).
    - What we meant is that 75 of the 209 Tolaki forms (35.9 percent) carry a cognate-set number. The list is not short of data: it fills 200 of 210 meanings, and 88.4 percent of its uncoded forms whose meaning the five Tolaki dialect lists cover have a look-alike in those lists (7.5 percent by chance).
    - What is sparse is the coding of the subgroup: by meaning, on shared meanings, the Tolaki list is coded for 38.3 percent, the dialect lists for 36.1 to 40.6 percent, and the 42 other Bungku-Tolaki lists for a median of 46.7 percent (five at or below the Tolaki figure); the unit matters because the share by form falls with the length of a list.
- **R1-8 — Muna *dh* is missing from the digraph list.** Two Muna forms contain *dh*: *akaradhaa* 'to work' and *idho* 'green'. Both are cognate-coded and both were converted in the test; the sentence omitted *dh* (and Wolio *gh*, which occurs in no form). The count of converted forms (75: Muna 54, Tolaki 20, Sa'dan Toraja 1) is unaffected. No loanword was removed from the classifier's data; *akaradhaa* carries ABVD's loan flag and stays in. The script's single-character placeholders for *gh*, *bh* and *dh* say nothing about Muna phonology, and the revision no longer gives phonetic values for the digraphs.
- **R1-9 — the glottal stop is written in several ways; same result under every convention?** Not quite, and the paper now says so (Section 3.6, Table 8).
    - The classifier sees only a written mark (ʔ or an apostrophe); the lists write it at very different rates (Makasar 36 percent of forms, Sa'dan Toraja 22, Bugis 21, Tolaki 14 — all with an apostrophe, and all 29 of them uncoded — Wolio four forms, Muna none), and whether these differences are notation or phonology is left to specialists.
    - Re-coding the mark as a consonant letter costs 0.016 of cross-validated AUC (0.020 to 0.024 for the held-out lists), leaving it unwritten 0.006, writing the pre-glottalised consonant as a geminate 0.007; unifying the apostrophe to ʔ changes no input. The AUC stays at or above 0.710 in every variant (held-out lists down to 0.677). By the rule written before the run, the result depends partly on the convention. The abstract's sentence that the profile is "robust … rather than orthographic artifacts" is withdrawn.
    - On the *k* of some South Sulawesi sources: eleven Sa'dan Toraja forms end in *k* beside 45 that end in a glottal mark in the same source, and since ABVD's PMP entries for two of those meanings end in *k*, we do not treat that *k* as a glottal stop; the text says so. Glottal mark plus consonant occurs in 35 forms and is also counted by the cluster input.
- **R1-10 — a semantic feature cannot belong to a phonological fingerprint.** Accepted, and the point goes further than we had seen.
    - Of the 26 inputs of the submitted headline model (the one behind the AUC of 0.763, after the coverage variable had been taken out), 17 came from the written form, 8 from the meaning and one was the identity of the source list. That input was the strongest of the 26. The submitted text did not show this: its figure and its ranking were those of the 27-input model, in which the coverage variable came first.
    - The model of the revision has no list input; its AUC is 0.727 with form and meaning inputs and 0.672 with the form inputs alone (meaning alone 0.646; list identity alone 0.680). "Trained exclusively on phonological features" and "phonological fingerprint" are withdrawn. The term is now *profile*, defined in Section 1 as part orthographic and part semantic, and the title has changed accordingly.
- **R1-11 — example lexemes for each quadrant.** Table 7 gives two forms for each of the four cells of Table 6, printed as ABVD records them with the cognate-set number, the PMP entry of

the meaning and the nearest same-meaning form in another ABVD list from the Sulawesi region, with its distance (above 0.34 it is not a look-alike by the screen). The text states that they have not been assessed etymologically and that, being the forms the classifier scores most extremely, they illustrate its inputs by construction. The full pool of 92 example rows (73 forms) and the scores of all 1,357 forms are in the released files.

- **R1-12 — "false positive" is used in two senses; and is the first of them not a false negative?** Yes: with "positive" meaning Austronesian, an inherited form left in the residual is a false negative, and the submitted text called it a false positive throughout.
    - The confusion came from our two label sets and from a "rescue" step that was in fact a concept filter (Part C). The revision uses one label with two values, *coded* and *uncoded*; "positive", "false positive" and "unlabeled positive" are no longer used as names of classes (the word "positive" remains only in "positive control" and in "positive or negative").
    - We have taken up the reviewer's remark that loanwords are commonly adapted to the borrower's phonology (Blust 2012:556 is cited for it): Section 4 says that an uncoded form that looks like the coded vocabulary is not thereby inherited and deserves a specialist's attention as much as one that does not.
- **R1-13 — what "documentation gaps" means for Tolaki; release the residual items.**
    - (a) We meant the second reading, cognates that exist but are not assigned in the database, and the phrase was badly chosen. On the first reading the data say only this: the list fills 200 of the 210 meanings, and of its 86 uncoded forms whose meaning the five Tolaki dialect lists also cover, 88.4 percent have a look-alike in those lists (7.5 percent by chance); whether a mainstream synonym exists that no list records, the study cannot show. The second reading is visible for part of the forms: of the 34 uncoded Tolaki forms that match an entry of the Proto-Bungku-Tolaki list, six match an entry that ABVD does assign to a cognate set, and twelve of the 128 uncoded forms whose meaning has a PMP entry resemble that entry by the screen (about two expected by chance; one is loan-flagged). For the rest the study cannot decide between a coding gap, a chance resemblance and a loan, and Section 3.3 says so.
    - (b) All forms are released, not only Tolaki's: one file with the 1,357 forms (ABVD identifiers, forms as recorded, cognate numbers and loan flags, the label, three out-of-sample scores, the nearest same-meaning forms with their distances), one with the 438 uncoded forms, and three smaller files (the 70 uncoded Makasar meanings with their relatives' forms, the twelve Tolaki look-alikes of PMP entries, the example pool behind Table 7), with a column description, deposited at Zenodo under CC BY 4.0. The DOI is given to the editors in the cover letter and withheld here because the record names the authors. The Tolaki subset has 134 items rather than the 114 of the submitted Table 1 because one label definition is now used (Part C).

Part B. Reviewer 2

We have rewritten Sections 2 and 3 for readers who do not work with classifiers: the terms the reviewer marked are explained in words, and the internal experiment codes are gone. The bullets follow the 24 highlights in the order of the pages of the reviewer's copy.

- **R2-1 — a reference for the claim that vocabulary "resists reconstruction".** We found no source that states the claim as we had written it, nor one that states it for Austronesian languages in general. The sentence now says only that in many Austronesian languages some forms cannot be assigned to a higher-level proto-form; it has no reference of its own, and the sentences that follow give what the literature does state: Reid (1994), who counted forms with no known cognate in three Philippine Negrito languages, and the literature on Makasar. The stronger triple claim of the submitted version (no cognates, non-canonical structure, irregular correspondences) is dropped.
- **R2-2 — what are E027 / E028 / E022?** Internal experiment codes; they appeared 25 times in the text and in the titles of two figures. They are gone from the text, which speaks of the label, the classifier, the screen and the permutation test. (The data statement names the folders of the code repository that hold the scripts, "experiments/E227 to E231"; these are folder names, not labels used in the text.)
- **R2-3 — the source of the semantic domains.** The seven domains and the "core" set are groupings made by the authors for this study, not Concepticon or the World Loanword Database; Section 2.3 says so, and it no longer calls the core set "Swadesh-100" (it covers 174 of the 210 meanings).
- **R2-4 — the "language identity" feature.** It was an integer numbering the six lists, and in the submitted headline model (26 inputs) it was the strongest input (see R1-10). It has been removed, and Section 2.3 says so.
- **R2-5 — what the classifier is and what "robustness" means.** Section 2.4 now opens with a sentence that says in plain words what a classifier is, and goes on to say what its score means; "robust" is no longer used. Three classifiers give the same picture (Table 3), which is all the submitted "robustness" amounted to.
- **R2-6 — implemented in scikit-learn / XGBoost — Python?** Yes; Section 2.4 gives the language and the version of each library.
- **R2-7 — what the false positives stand for.** The term is no longer used (see R1-12).
- **R2-8 — the "E022 binary label" cannot be found in the rule-based section.** Because the label the models used was not the one that section described. The revision has one label, defined in Section 1 and, with what follows from it, in Section 2.2.
- **R2-9 — kappa: agreement?** Section 3.5 defines it where Table 6 reports it (Cohen's kappa between the label and the model's out-of-sample score at a threshold of 0.5; 0.31). The submitted 0.61 was computed on the training data and is withdrawn.
- **R2-10 — why k = 5 to 30?** There was no reason; the clustering is removed (Part C).
- **R2-11 — "both models" — which?** The first of them (31 inputs, four of which encoded the label) is removed; one model family remains, with a form-only variant beside it.
- **R2-12 — "near-perfect" and how to read Table 2.** The near-perfect figure belonged to the removed model. The new Table 3 gives, for each classifier, the AUC, the accuracy beside the accuracy of calling every form coded (0.719 against 0.677), and the precision, recall and F1 of the uncoded class (the submitted F1 of 0.82 was the coded class's). Section 2.5 explains each column in words, and Section 3.4 reads the first row of the table.

- **R2-13 — AUC: what does it stand for and mean?** Defined in Section 2.5: the probability that a randomly chosen uncoded form outranks a randomly chosen coded one; 0.5 is chance, 1.0 is perfect.
- **R2-14 — which range counts as moderate or reliable?** Section 2.5 says that verbal scales exist and gives one (Nahm 2022, Table 4), but that they are conventions without an empirical basis (White et al. 2023), and the paper reports numbers rather than labels. The "0.65 threshold" of the submitted text was our own working line and is dropped.
- **R2-15 — what "generalises across Sulawesi languages" means.** Section 2.5 defines it (a model trained on five lists ranks the forms of the sixth) and Section 3.4 reports it: AUC 0.614 to 0.809 for the held-out list, mean 0.701, and — new — that at a threshold of 0.5 the classifier is below the majority answer for five of the six lists. The ranking carries over to an unseen list; the classification does not.
- **R2-16 — "feature ablation" and the variable in typewriter font.** The word "ablation" is no longer used; where the classifier is run again without a group of inputs, the text says so in those words (Sections 2.6 and 3.6). The variable in question no longer exists — its values in the code were stale constants, not the coverage they were named for — and nothing is printed in typewriter font; where the code's own name for an input says more than the input measures, Section 2.3 says so.
- **R2-17 — SHAP.** Section 2.4 explains what the bars and the arrows of Figure 1 measure; Section 3.5 says that the figure describes the classifier and not the vocabulary.
- **R2-18 — CV.** Cross-validation is explained in words in Section 2.5 (five parts, ten repetitions, fifty test sets).
- **R2-19 — the symbol Δ.** It now occurs only as a column heading of Table 8, and the note to the table says what it is: the change in AUC from the first row.
- **R2-20 — the panels of Figure 2 are neither referred to nor explained.** The submitted Figure 2 is removed; Table 6 replaces it, with examples in Table 7. (Figure 2 of the revision is the new workflow diagram.)
- **R2-21 — why 'One Hundred', 'Fifty', 'Twenty', 'to stand' and 'to hit'?** Section 3.5 answers for four of the five: 'Fifty' and 'Twenty' appear to be decimal compounds whose parts recur across the lists (uncoded in four, coded in Bugis); 'to stand' and 'to hit' were on our earlier list of inherited meanings, and the "consensus" had been computed on a label that ignored that list. 'One Hundred' is likewise uncoded in four lists and coded in Bugis; the text prints its five forms as ABVD records them and gives no account of them. Out of sample, no meaning is "uncoded, profile" in four or more lists.
- **R2-22 — DBSCAN parameters.** Removed with the clustering.
- **R2-23 — Table 5 is cut off.** The submitted Table 5 and Figure 4 are removed; Section 3.8 reports the sixteen-list application in one paragraph without a geographic claim (Part C).
- **R2-24 — define "fingerprint" at the start.** The term is now *profile*, defined in Section 1; because its inputs include the meaning, it is not called phonological.

We have kept the three things the reviewer singled out for approval: the limitations (a paragraph of Section 4, rewritten for what the revision claims; limits are also stated with each result), the declaration on the use of artificial intelligence (rewritten to name each stage at which the assistance was used), and the open code and data (now with a Zenodo deposit). The reviewer also found the Discussion easier to follow than Section 3, and its insights of potential interest. Much of the submitted Discussion rested on results that did not survive our re-derivation (Part C) and is withdrawn; the revised Section 4 is shorter and says what the measurements do and do not show for the profile, for Makasar and for Tolaki.

Part C. Corrections by the authors

Found when every number of the submitted text was re-derived from the raw files before this revision; not raised as such by the reviewers, though several of their questions lead to them. The stored result files reproduce from the data; the manuscript text did not describe them correctly.

- **Two residual sets presented as one.** Submitted: 356 forms in Table 1; 438 in every model and in the abstract as "26.5 %". Revised: one label: 438 uncoded forms = 32.3 percent; per list 34 / 62 / 80 / 83 / 45 / 134 (Table 1).
- **"Proto-Austronesian cross-check".** Submitted: described as a comparison of residual forms with reconstructions. Revised: it compared no forms (all uncoded forms for 15 meanings were removed); step deleted.
- **Loanword filter.** Submitted: Sanskrit, Arabic and Malay trade-word patterns. Revised: matched five unrelated words; deleted.
- **Meanings residual "in five or more of the six languages".** Submitted: eight meanings, presented as the strongest candidates for pre-Austronesian substrate or shared innovation. Revised: the count was of forms, so a list with two or three synonyms for a meaning counted two or three times; by lists, three of the eight qualify ('if', 'to throw', 'to tie up, fasten'); the list is dropped.
- **"Fewer canonical Austronesian prefixes".** Submitted: abstract and three places in the text. Revised: the onset-string input is more frequent among uncoded forms (37.0 vs 25.2 %) but not separable from length; neither "fewer" nor "more" is claimed (Section 3.5).
- **Headline AUC 0.763 "phonological only".** Submitted: 26 inputs including the identity of the source list. Revised: form + meaning 0.727 (held-out list 0.701); form only 0.672 (0.641); no list input.
- **"Two-method consensus", $\kappa = 0.61$, 266 forms.** Submitted: agreement computed on the training data. Revised: out of sample $\kappa = 0.31$; cells 172 / 266 / 105 / 814 (Table 6); the consensus framing removed. (That 266 recurs is a coincidence: on the training data 266 of the 438 uncoded forms scored 0.5 or more; out of sample 172 do and 266 do not.)
- **$p = 0.569$; "optimal k = 30".** Submitted: a test that did not test the hypothesis; the edge of the search range. Revised: permutation test: a small excess of same-meaning similarity among uncoded forms (Section 3.7); clustering removed; no claim about a shared layer either way.
- **Sixteen-language application.** Submitted: mean score 0.606 vs 0.393 "significantly"; Table 5 with four stale constants. Revised: two list-level inputs drove the pattern; without them mean AUC 0.638 and no geographic claim (Section 3.8); the submitted Table 5 and Figure 4 removed.
- **Robustness section.** Submitted: 0.772 against 0.774 and similar pairs; "does not depend on form length at all". Revised: the submitted numbers were not those of the model that section named: they came from the 26-input model under another division into folds. For the models of the revision (Section 3.6), digraph conversion changes nothing (75 forms) and removing both size inputs costs 0.014 / 0.023 AUC; the sentence is withdrawn.
- **Figure 1 caption; Table 2 F1; class weighting.** Submitted: sign reversed; F1 0.82 = coded class; XGBoost described as adjusted for class imbalance. Revised: Figure 1 redrawn; the F1 of the uncoded class is 0.480, and the accuracy is 0.719 beside 0.677 for calling every form coded (Table 3); XGBoost was run without class weighting, and Section 2.4 says so.
- **"Swadesh-100" flag; coverage variable; "syllable".** Submitted: a flag that covers 174 of the 210 meanings; a variable that held stale constants; "syllables" that were groups of vowel letters. Revised: the core set is described as what it is (Section 2.3); the coverage variable and the syllable count are no longer used.

- **The submitted Section 4.5 (Javanese script).** Submitted: “convergent evidence”. Revised: removed: not asked for, outside Sulawesi, resting on properties that changed, and quoting a percentage for the wrong group.
- **References.** Submitted: Ross 2005 given as an article in this journal (it is a book chapter); six further entries of the reference list with a wrong entry type, publisher or pages. Revised: every entry of the reference list was checked against the publication, its catalogue record or, for a few page ranges, the reference list of another work; Ross 2005 is no longer cited; the list is set in the journal’s style.